Observing Crossing Over

Biology 2e chapter 11 reading connection | Teacher guide / classroom preparation

Learning target

What can ordered spore patterns reveal about crossing over?

Timing and materials

One class with prepared specimens; live culture preparation is an optional advanced route.

Prepared Sordaria cross slide with intact ordered asci or suitable educational cross kit, compound microscope, scoring sheet. Use prepared material for the basic route.

Classroom route

1. Inspect the prepared cross at low power, then use sufficient magnification to distinguish the eight spores. Follow supplier preparation instructions if using a kit.

2. Choose fields without selecting for an interesting pattern. Count intact eight-spore asci containing four spores of each color; record excluded or ambiguous asci separately.

3. Classify 4:4, 2:4:2 and 2:2:2:2 arrangements in at least two non-overlapping fields. Aim for a larger pooled sample than the virtual demonstration. Estimate gene-to-centromere distance as 0.5 x percent second-division segregation.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

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Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

Expected reasoning

4:4 is first-division segregation. 2:4:2 and 2:2:2:2 are second-division patterns. The simple distance estimate is half the second-division percentage because half the chromatids are recombinant in a single crossover. Multiple crossovers complicate interpretation.

Feedback and scoring

Criterion	Points	Evidence
Prediction and controls	0-2	Testable claim; comparable samples and controlled variables.
Observations	0-2	Complete, labeled records with units and source.
Interpretation	0-2	Correct calculations or categorical comparisons; claim tied to evidence.
Limitations	0-2	Specific uncertainty and a useful follow-up investigation.

Score reasoning and data quality, not agreement of physical results with the simulation. Manual tallies are student observations; a matching total does not prove every category is correct. Do not replace student counts with the model answer. The optional Sordaria extension retains classification feedback.

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Model limitations

Original ordered-spore schematics. The simple mapping estimate assumes the scoring model and can miss multiple crossovers; this is gene-to-centromere mapping, not ordering several genes.

Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Corn and fly activities require manual tallies summing to the sample size plus review; optional specimen marks do not classify or tally automatically. Chromosome models require all four gametes plus a reviewed tally; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

Evidence and optional classroom resources

Suitable Sordaria specimens

https://www.carolina.com/fungi/sordaria-genetics-class-biokit/FAM_155847.pr

External resources remain on their publishers' websites. This packet contains original text and tables, not supplier worksheets. Chapter numbers are reading connections. No supplier or textbook publisher endorsement is implied.

Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.